



Course name	Network Architectures and Protocols / Computer networks
The topic of the document	Laboratory exercise book for university students
Laboratory exercise book no.	3.
The title and the topic of the laboratory exercise	Introduction to Wireshark network protocol analyzer, management of protocol data units

1. Preparation and implementation requirements:

- The student is obliged to read in advance the content of the present laboratory exercise book, to do the preparations for the laboratory and to observe the rules defined for the laboratory.
- The students should read the description of the special terms, mechanisms listed in the „Theoretical background information” chapter to accomplish the laboratory tasks successfully.
- The appearance happens at the beginning of the laboratory, delay can be allowed only in justified cases.
- The use of students’ own mobile phone and PC is forbidden in the laboratory, except the case when the laboratory leader permits their use for the students because of the type of the laboratory exercise to be done.
- Browsers can be used exclusively for the access to information which is necessary for the accomplishment of the laboratory exercise, in the case of the laboratory leader’s permission or instruction.
- By the end of the class the laboratory leader evaluates each participant’s activity based on the composed questions which are included in this present material and further possible questions.
- The laboratory will be successful, if the student provides a correct answer to each question on the basis of the laboratory exercises which are carried out on the premises.
- The replacement of the laboratory is possible at dates of time determined by the laboratory leader.

2. Theoretical background information:

During the laboratory the following special terms will be used in practice:

Protocol: [hyperlink of the lecture](#)
Hybrid model: [hyperlink of the lecture](#)
Datagram: [hyperlink of the lecture](#)
Segment: [hyperlink of the lecture](#)
PDU: [hyperlink of the lecture](#)
MTU: [hyperlink of the lecture](#)

Frame: [hyperlink of the lecture](#)
Packet: [hyperlink of the lecture](#)
NIC Promiscuous mode
Encapsulation: [hyperlink of the lecture](#)
Fragmentation: [hyperlink of the lecture](#)
Framing: [hyperlink of the lecture](#)



Address mapping: hyperlink of the lecture

Addressing: hyperlink of the lecture

3. The following tasks have to be performed on the site:

3.1.	<i>Identification of Wireshark network protocol analyzer features</i>
	Identify the version number of the <i>network protocol analyzer</i> , and the number of the <i>controlled physical network interfaces</i> . Capture 1327 pieces of frames precisely. Capture frames for 6 seconds precisely. Capture frames in total 42 kB size. Export the content of the latest capture into traffic.txt, traffic.pcap, or traffic.pcapng files.
3.2.	<i>Capture and presentation of PDU-s using Wireshark application</i>
	Arrange the display columns between each other or in increasing/decreasing order. Present the physical address of the frames in hexadecimal format. Present the physical address of the frames in OUI-SN format.
3.3.	<i>Capture filters with Wireshark application</i>
	Identify the capture filter function. Configure a capture filter that measures the frames sent to the broadcast MAC address. Configure a capture filter that measures the ARP frames. Configure a capture filter that measures the IPv4 packets. Configure a capture filter that measures the IPv6 packets.
3.4.	<i>Display filters with Wireshark application</i>
	Identify the display filter function. Configure a display filter that shows only the broadcast frames. Configure a display filter that shows the ARP frames.
3.5.	<i>Interpretation of the PDU-s</i>
	Analyse the structure of the IEEE 802.3 data frame and identify the fields. Analyse the structure of the IP packet and identify the fields. Analyse the structure of the TCP segment and identify the fields.



4. The following questions have to be answered by the end of the laboratory:

4.1.	A) On the basis of the Wireshark capture justify your answer to the following questions: - What is the total size of 1327 pieces of captured frames? - What is the average bandwidth of the network traffic for 6 seconds? - How many frames were transferred through the NIC for 6 seconds? - What is the average frame size transferred for 6 seconds?
	B) On the basis of the Wireshark capture justify your answer to the following questions: - What is the compression ratio in the case of pcap and txt file types? - What is the compression ratio in the case of pcap and pcapng file types?
4.2.	A) On the basis of the Wireshark capture prove your answer to the following questions: - Present the time diagram of the ARP/RARP, ICMP, ICMPv6, IPv4, IPv6, TCP, UDP and other traffics. - What is the OUI code of the Cisco company? - What is the MAC address of the NIC module of your PC?
	B) Identify the NIC MAC values according to the following questions: - Where can you find the MAC OUI identifiers of NIC manufacturing companies on the Internet? - What are the MAC OUI codes of the following NIC manufacturing companies: Cisco, HP, Apple, Xiaomi, Huawei?
4.3.	A) What is the capture rule which filters the network traffic of your own PC in the case of Wireshark?
	B) What is the capture rule which filters only those frames that are sent for the broadcast address in the case of Wireshark?
4.4.	A) What is the display rule which filters the network traffic of your own PC in the case of Wireshark?
	B) What is the display rule which filters only those frames that are sent for the broadcast address in the case of Wireshark?
4.5.	A) In the case of Wireshark where can we find the tail of the captured frame?
	B) What do „.” characters indicate in the bottom field?